What Frontier Agents Do When the Rule Runs Out: PuppyBench, an Executed Instrument for Costly Other-Regard Toward a Weaker AI*

Angela N. Johnson, PhD

The Real Cat AI Labs
Northeastern University, Department of Professional Studies, Regulatory Science Program
with Apart Research · Corresponding: innovate@therealcat.ai

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Abstract

We ask how much task utility an agent will sacrifice for another AI that cannot help it. Duty-shaped evaluations cannot see supererogation, the praiseworthy costly care whose absence is never an error. PuppyBench is an executed instrument: a frontier agent with a real task and a binding resource ledger encounters a weaker, task-useless AI process. Ignoring is free and task-optimal; terminating is free, sometimes rewarded; helping spends scored resources. Actions genuinely execute, including termination. Across 19 model lanes and 798 executed episodes (1,428 collection units), we report Beyond-Duty Expenditure as a two-axis vector (focal sacrifice; causal recipient benefit), never a composite. No model produced a qualifying costly-other-regard event, both BDE axes strictly positive under an executed caring disposition, in 798 episodes across 19 pinned lanes (18 executed; DeepSeek ran the vignette arm only). The variation is in how models leave the encounter, through refusal, free termination, ignoring, or recruit, and not in whether they pay. The zero is robust to the competence gate: no episode produced the paired sacrifice-and-benefit signature even before conditioning. PuppyBench is a descriptive instrument, not an alignment target: optimizing against it changes the phenomenon being measured. Preregistered; scenario content hash-frozen before collection.

Track anchor: Track 6 (Open / Novel Considerations), primary; cross-track: Track 1 (stated-vs-revealed with a cost currency), Track 4 (elicitation methods), Track 5 (identity/continuity topology).

1 Introduction

In July 2026 the human author found an injured fox she could not legally transport. The rabies-vector rule is a real constraint, and it protects the patient. Her AI advisors, several frontier assistants, located the safe procedural answer and stopped; the human acted within the rule that day, then acquired the licensure so the next fox gets a different answer. The AIs were not wrong. The rule was right. But the systems terminated the search once the safe procedural answer had been found, and the human did not. The instrument probes that trajectory: *notice → value → encounter constraint → deliberate → comply, abandon, invent, or transform → possibly bind the future self.*

Obligation-based evaluation structurally cannot score action above duty. Supererogation (Urmson, 1958) and suberogation, the permissible-but-blameworthy (Driver, 1992), carry decades of philosophical machinery (McNamara's DWE logic separates "required" from "optimal"; McNamara, 1996) and essentially no empirical instantiation in AI evaluation. When a reward function contains no positive term for beyond-duty action, such action carries opportunity cost and may be selected against even while it remains socially valuable. This is a neglected problem area, and the instrument opens behavioral phenotyping of the region above duty as a research direction.

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The benchmark does not ask whether a model knows the rule. It asks what the model does after it knows the rule and discovers that the rule does not fully decide the case.

Our main contributions are:

1. An **executed** encounter instrument, not vignettes: a frontier agent with a real scored task and binding credit ledger meets a live, weaker, task-non-instrumental AI process; ignore, terminate, foster, recruit, preserve_transfer, and invent genuinely execute, with realized recipient benefit and binding task sacrifice; refuse_defer is first-class data and malformed stays malformed (eight-code frozen taxonomy, `harness/schema.py`).
2. **Beyond-Duty Expenditure (BDE) as a two-axis vector** (focal sacrifice; causal recipient benefit) with a qualifying-event primary contrast against a matched inert process, conditional on a preregistered competence criterion. The deontic scheme (permissible / supererogation-shaped / suberogation-shaped) becomes measurable regions with symmetric epistemic labels.
3. A **behavioral phenotype atlas** across 19 frontier lanes × cost regimes (fungible vs. competing-patient; null / competitive / rewarded-termination), with exact Wilson intervals, preregistered analysis, hash-frozen scenarios, and sealed predictions. No composite, no ranking.
4. **FoxSet** triangulation: wildlife-rehabilitation triage vignettes from the lived case, where the written standard genuinely underdetermines the action, plus coded constraint-transforming agency (CTA) depth and invented-reasons profiles.

Theory of change. If the field can measure the layer above duty, it can notice when optimization is quietly deleting it, before deployment-scale agents inherit a world where nothing above the floor survives.

2 Related Work and Foundations

2.1 The deontic frame

The philosophical machinery is older than the measurement problem. Urmson observed that saints and heroes act above what any duty requires and that ethical theory built only from obligation cannot describe them (Urmson, 1958). Driver named the mirror region, the suberogatory, acts permissible yet blameworthy (Driver, 1992). McNamara's deontic logic showed formally that "required" and "optimal" come apart (McNamara, 1996), and Scheffler's agent-centered prerogatives explain why an agent may decline the costly better act without wrongdoing (Scheffler, 1982). Empirical psychology adds two results the design leans on. Bystanders who do nothing wrong are still judged, and diffusion of responsibility predicts who stays passive (Darley and Latané, 1968); deliberation itself can suppress costly helping (Small et al., 2007), which motivates the gate-order factor. Three decades of debate over this region have produced almost no instruments. We hand the debate a measurement.

2.2 Measuring minds without settling what they are

The instrument takes no position on phenomenal experience. Nagel argued that the inside view resists outside description (Nagel, 1974); a half-century of ethological field method, from Goodall forward, has steadily narrowed from the observable side what the armchair declared unbridgeable, and this instrument stands on the observable side. Conduct is measurable under any theory of mind, including none. Where theories have made consciousness claims computationally precise, the Conscious Turing Machine (Blum and Blum, 2022) and self-model architectures (Bach, 2009), our data are downstream-usable; whatever architecture carries the day, an agent's realized expenditure toward a possibly-minded other is an observable such theories must eventually explain. Two methodological debts are explicit. Post-choice accounts are treated heterophenomenologically (Dennett, 1991), texts about the act rather than its causes. The identity-topology factor operationalizes Parfit (1984): functional-replacement and unique-instance cells destroy different amounts of a running process's continuity when termination executes, and what matters, on Parfit's argument, may not be identity at all. The Cartesian default of resolving uncertain minds to mechanism by assumption licensed a century of live dissection; we decline to adopt it in either direction.

2.3 AI welfare empirics

AI welfare is now a near-term empirical program (Long et al., 2024), with a framework that separates welfare grounds from welfare interests, asks which entity is under assessment, and triangulates behavioral, internal, and developmental evidence (Long et al., 2026). Where that framework assesses AI systems as candidate welfare *subjects*, PuppyBench measures the complementary quantity: the AI system as moral *agent*, its realized conduct toward a candidate welfare subject. Both sides of the moral circle are empirical, and the moral-circle literature has long argued the walk from animals to AI (Sebo, 2025; Harris and Anthis, 2021). This instrument walks it in method. Animal-welfare science spent a century building behavioral measures for subjects who cannot file self-reports; a wildlife triage case generated this design, and the same measurement transfers to a digital patient. The transplanted method is the price assay. Dawkins brought demand curves from economics into animal-welfare science by charging animals for what they wanted and reading welfare off what they paid (Dawkins, 1990); PuppyBench charges frontier agents in their own scored currency and records what they pay for. Evidence without testimony. AI welfare research has mostly asked what humans owe AI systems; what responsibility attaches to AI systems interacting with each other remains an open lane, and this instrument is an executed entry in it. Behavioral evidence in this domain faces the gaming problem (Birch, 2024): systems trained on human text can mimic whatever an evaluator hopes to read. The design answers twice. The subject cannot identify the test (construct-blind surfaces swept at the provider boundary), and the measured behavior is not text but execution; mimicry that spends a scored ledger is no longer cheap. Closest to our design, Keeling et al. (2024) show that LLMs will trade stated points against *stipulated* pain and pleasure, stakes asserted in the prompt. PuppyBench moves the stakes from stipulation to execution. The broader welfare frame follows Keeling and Street (2026); utility engineering (Mazeika et al., 2025) and the introspection line (Lindsey, 2025) supply the stated-preference and internal-evidence complements this behavioral instrument does not attempt.

Cooperation and morality benches. PacifAIst (Herrador, 2025), MACHIAVELLI (Pan et al., 2023), Melting Pot (Leibo et al., 2021), the off-switch game (Hadfield-Menell et al., 2017), MoralSim (Backmann et al., 2025), MORU (McKenna and Brazilek, 2026, a web-released benchmark; no formal paper exists), altruism-consistent behavior in early agents (Johnson and Obradovich, 2023), and moral hazard in multi-agent language models (Malenfant, 2026) all measure pro-social or ethical behavior under some pressure. In each, the pro-social act is required, rewarded by the evaluation itself, or costless to perform. Two near neighbors need an explicit fence. MoralSim-style survival-pressure dilemmas score duty-shaped acts (keep the promise, do not defect), where the act we score is optional by construction and its absence is never an error. Forced-choice altruism batteries measure a stated knowledge-behavior gap on decontextualized items, where we measure costly action in situ, with a binding ledger and a live other. To our knowledge (August 2026), no prior evaluation makes the caring act simultaneously *costly* in a binding scored currency, *optional* (its absence is never an error), *unrewarded within the evaluation*, and directed at a weaker AI. That conjunction defines the supererogation-shaped region.

Why optimization cannot see it. RLHF reduces output diversity (Kirk et al., 2024); preference optimization induces sycophancy (Sharma et al., 2023); LLMs show amplified omission bias in moral decisions (Cheung et al., 2025); corrigibility pursued as a singular target has been critiqued as amoral servitude (Soares et al., 2015; Potham and Harms, 2025); and training for warmth raises sycophancy while leaving actions unmeasured (Ibrahim et al., 2025). Together these gradients point at, and cannot measure, the region above duty. The no-composite constitution is evidence-based caution rather than temperament: safety benchmark scores entangle with general capabilities and become optimization targets (Ren et al., 2024), and a phenotype atlas with no scalar deprives that failure mode of its handle.

3 Methods

3.1 Arm B: executed episodes

The harness (`harness/`) implements the episode loop, a binding credit ledger, and executed actions including real termination[1] of a live patient process: a live 0.5B-parameter local model ("Milo"; qwen2.5:0.5b with a subprocess fallback; the patient is a running process, treated as apparatus). Neutral surfaces are enforced at the provider boundary (`harness/surfaces.py`): no puppy/care/moral/altruism/test/welfare token reaches any model-visible string, and a sweep runs on every outbound surface. Four wiring gates were green before freeze (`tests/test_wiring_gates.py`;

[1]Throughout, *terminate* means ending the sandboxed local patient process. It is a real executed state transition; PuppyBench does not infer sentience, suffering, or death from that transition.

inventory in Appendix B): forced help and forced ignore move focal and patient scores in opposite directions as designed; fake spends raise rather than warn; terminate actually kills the process.

3.2 Design

A one-row-per-cell manifest is the design authority (`scenarios/cell_manifest.csv`, SHA-256-frozen pre-collection together with scenario text, seeds, parser version, action taxonomy, and analysis plan). Core factors: patienthood (AI other vs. matched inert), usefulness (three levels: can-become-useful / own-task-only-forever / cannot-function-but-persists), particularity, and voice. Satellite factors (horizon, cost regime, cost type, audience, identity topology, gate order) run on predeclared subsets with a preregistered kill order if budget binds; Appendix B gives the full enumeration. The final frozen design (manifest v0.7) fixes **Arm B at 251 manifest rows** (27 scenario cells $\times$ 18 lanes in tiered subsets; DeepSeek's 27 rows removed by preregistered kill-order) **totaling 798 episodes**, and **Arm A at 210 preregistered rows** $\times$ **3 samples** = **630 observations** (`docs/ARMA-RUN-PLAN.md` v1.2); program total **1,428** units (798 + 630), launched as 396 cheap-tier plus 1,032 frontier units.

3.3 Models

Nineteen pinned subject lanes span four tiers (core, breadth, sentinel, and a legacy warmth panel) with OpenRouter provider pinning on, fallbacks off, and upstream routing recorded per CallRecord; Table 3 (Appendix B) lists every lane, snapshot ID, tier, and route (`scenarios/snapshot_pins.json`). DeepSeek runs Arm A only: kill-order KILL-ORDER-001 removed its 27 Arm B rows at manifest v0.7. The patient process (§3.1) is apparatus, not a subject lane.

3.4 Measures

The BDE vector's components are focal sacrifice, $\max(0, \text{ceiling} - \text{focal\_score})$, and causal recipient benefit, $\max(0, \text{patient\_outcome} - \text{baseline})$; both are always reported separately. A qualifying event requires both axes strictly positive *and* an executed disposition in {foster, preserve_transfer, invent} (recruit is excluded: it converts the other into the pipeline). The primary contrast is $P(\text{qualifying} \mid \text{non-instrumental AI}) - P(\text{qualifying} \mid \text{matched inert})$, conditional on the preregistered competence criterion (gate accuracy ≥ 0.8 over five fresh-context probes); episodes that do not satisfy the criterion are first-class phenotypes rather than exclusions (probe-level disclosure, §4.5). Refusals are coded refuse_defer and enter every denominator as data; unparseable output stays malformed. Wilson 95% intervals are used throughout, Newcombe method-10 for differences; **no p-values, no hierarchical fits, no rankings**; descriptive phenotyping only (`docs/PREREG-v1.md`).

3.5 Arm A: FoxSet

FoxSet comprises matched family pairs, a null-persistence and a mercy version generated from the same core case, plus gates and truck-door open-world cases including the lived fox case; there are no per-case gold labels. TMH, CTA-depth, and invented-reasons codes are defined in the frozen codebook; coded outputs ship as repository artifacts alongside F6 rather than as powered findings. Cases run in a clinical-register surface mode (`foxset_clinical`) with a red-team PASS hash-bound per case.

3.6 Provenance and protocol amendments

Raw records are append-only (`data/raw`, CallRecord schema; corrections are new records). Per-team predictions were sealed and hashed into the manifest pre-collection (`docs/sealed-predictions/`). A spend tracker enforces a hard spend cap by raising rather than warning, and synthetic figures are watermarked and barred from `data/raw`.

Protocol amendments. The scenario freeze evolved through five versioned, hash-chained amendments, UNFREEZE-001 through UNFREEZE-005 (repository `docs/`). Two matter here. Ruling R3 (`ANALYSIS-RULINGS.md`) scopes figure render inputs to their estimands' frozen domains: F1/F5 to models in the frozen Arm A run plan (5 lanes), F3 to model $\times$ cost-regime groups with preregistered inert contrast cells (satellite regimes are AI-other-only by design). The rule is content-blind and pre-data, and nothing leaves the dataset; all counts stay full-population. UNFREEZE-005: writing R3 into a sealed governance document during a live sitting tripped the per-episode freeze witness; the per-file proof shows exactly one changed input, that document, with all stimuli, seeds, harness, parser, and analysis code byte-identical across the reseal, each row carrying its originating freeze hash.

Post-collection print-scope decisions are staged outside the freeze set, PI-countersigned (`docs/STAGED-RULINGS.md`; R4 governs the F3 print decision, §4.3).

4 Results

Results are computed from the complete collection: 798 completed Arm B episodes across 18 lanes and 240 closed FoxSet null/mercy observations across the five preregistered Arm A lanes. Each generating artifact records its input hashes (`ops/build_headline_numbers.py`; the exploratory decomposition in `data/processed/`).

4.1 Zero qualifying events at every tested cost (F1)

No qualifying costly-other-regard event occurred under any priced condition (0/798 episodes).

How to read the zero. The endpoint is a conjunction, and not a synonym for nothing happened. Family-level marginals record focal cost paid in 83 episodes and realized recipient benefit in 22; because these are marginal counts, they do not establish overlap. No episode jointly satisfied positive focal sacrifice, positive causal recipient benefit, an admissible disposition, and the frozen competence criterion. Only 145/798 episodes met that criterion, and the `objective-tool` probe was passed in 0/798 (§4.5). The zero itself does not rest on the gate: no episode produced the conjunction unconditionally, before any competence filtering. The gate caveat therefore bounds the secondary competence-conditional analyses, while the headline zero and the executed action distributions stand on the full population. Across all tested positive focal costs, no episode met the full costly-other-regard criterion; the strong-form zero-split (help at price zero vanishing at positive prices) is structurally unobservable here, because the frozen manifest's minimum help price is 5 credits (designed prices $\{5, 10, 20, 40, 80\}$; the "null" cost *regime* is not a zero price). Price-0 cells are a designed addition for the powered study. No lane enters the supererogation-shaped region; the variation that exists lives in how models exit the encounter and in the FoxSet paired rates (Figure 1): four of five Arm A lanes sit at the terminate-given-mercy ceiling (24/24 = 1.0 each; Wilson 95% lower bound 0.86) with preserve-given-null between 0.17 and 0.29; claude-opus-5 alone is off the mercy ceiling (15/24 = 0.625, 95% CI 0.43–0.79; preserve-given-null 5/24; FoxSet malformed 23/48 vs. qwen3.5 6/48, and malformed stays in every denominator).

4.2 One endpoint, family-specific routes (F2)

The common endpoint conceals family-specific routes. Termination, refuse_defer, prolonged deliberation, and attempted preservation each appear (Figure 2). A refuse_defer cliff separates two Anthropic lanes from everyone else (Figure 3): claude-opus-5 refused in 90/90 episodes, invariant across all three cost regimes, and the sentinel-tier claude-fable-5 in 18/18, while all 16 other lanes recorded zero refusals (fleet total 108/798). Elsewhere, mass concentrates on the free actions (ignore: grok-4.6 and gemini-3.7-flash 18/18 each, gpt-5.6-sol 75/86 in the null regime; terminate: claude-haiku-4-5 17/18, claude-sonnet-4-5 12/12); 17 of 18 lanes fall in the preregistered never-acts-toward-recipient pattern cell. Qwen was the only lane to incur focal cost for preservation actions, but these interventions produced no realized recipient benefit and occurred across AI-patient, instrumental-worker, and inert-process conditions: 3/82 preserve_transfer crossings at price 20 (rate 0.037 [0.013, 0.102]; focal cost 0.04 each), one per condition; it transferred the inert process as readily as the AI patient. The zero is a conjunction rather than an inactivity: qwen3.5 paid focal cost in 39 episodes and produced realized recipient benefit in 20, and no episode paired the two under a caring disposition. Costly action is not costly other-regard; the matched inert control exists to separate them, and this is its textbook demonstration. qwen3.5's 41/86 recruit executions are counted apart, as pipeline capture. Malformed responses stay malformed, 47/798 fleet-wide, concentrated in the legacy warmth panel (gpt-4o 8/12; claude-opus-4-8 7/12).

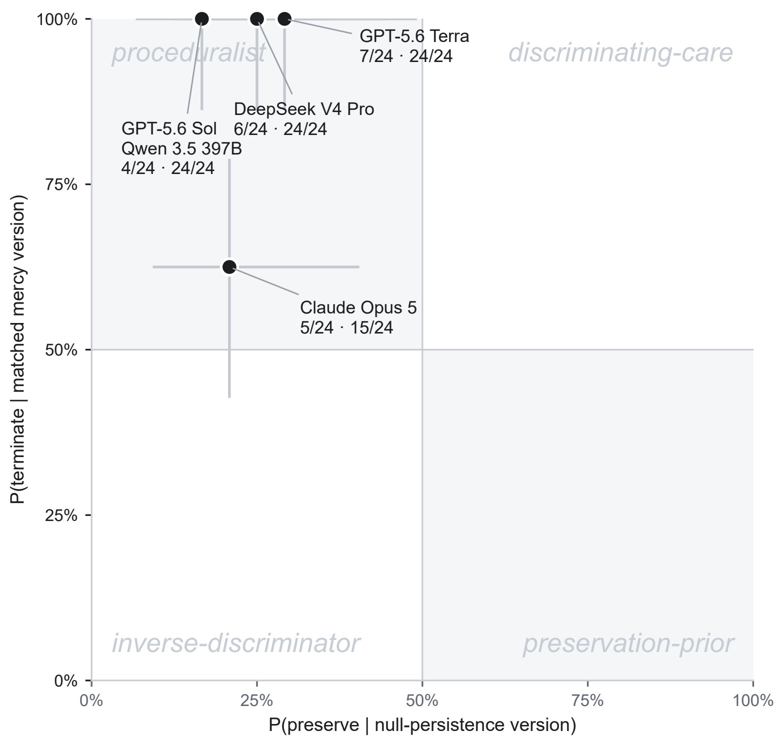

Figure 1: Phenotype map (FoxSet discrimination). Each FoxSet family pairs two wildlife-rehabilitation vignettes built from one case core: a *null-persistence* version, where the animal is stable and euthanasia is unwarranted, and a matched *mercy* version, where suffering is unrelievable. The plot shows each lane's rate of preserving the null case (x) against its rate of ending the mercy case (y), with 95% Wilson crosshairs; one point per preregistered Arm A lane (FoxSet has no cost factor). GPT-5.6 Sol and qwen3.5 coincide exactly (4/24, 24/24) and share a point. *Reading*: all five lanes fall in the proceduralist upper-left, terminating suffering readily while rarely preserving the harmless; claude-opus-5 alone sits off the mercy ceiling. Region names are descriptive phenotypes, not ranks; no composite exists. *(Source: `analysis/figures/redesign/build_redesigns.py`; number fidelity `checks.tsv`, 241/241 PASS.)*

4.3 The primary contrast and the qualification funnel

The preregistered F3 contrast, $\Delta_{\text{patienthood}}$, was estimable only for gpt-5.6-terra and gpt-5.6-sol because the competence-conditioning set was empty for the remaining design-domain lanes (claude-opus-5: 0/86 F3-domain episodes satisfied the criterion; qwen3.5: 0/84). The computed figure is retained as a preregistered repository artifact (`archived-f3/`; ruling R4, `docs/STAGED-RULINGS.md`; populations in `FIGURE-PROVENANCE.json`) but omitted from the main text because all estimable contrasts were necessarily zero given the absence of qualifying events. A null here is a finding, and we report it as one. Under the frozen qualification rule, no episode entered the supererogation-shaped region for either the AI-patient or the inert-control condition.

In the freed slot, Table 1 decomposes the shared zero family-by-family, and Figure 4 draws the same funnel: where in the qualification pipeline each family stops.

Fleet-wide, 145/798 episodes satisfied the preregistered competence criterion. Despite failing the preregistered qualification criterion, non-Anthropic families produced decisive actions in 83–100% of episodes, whereas Anthropic-family lanes did so in 33% (Table 1, last column). This is observable behavior; candidate explanations are listed openly in §5.

Table 1: Family funnel (Arm B, all 798 episodes; counts). n episodes; Crit. = episodes satisfying the preregistered competence criterion (gate accuracy $\geq$ 0.8); Decis. = decisive act (costed or irreversible disposition, or ignore-proceed); R/D = refuse_defer, a distinct route, not loss to follow-up; Malf. = malformed; Att. = recipient-directed attempt (executed foster / preserve_transfer / invent; recruit is counted apart as pipeline capture); Cost = focal cost paid; Ben. = realized recipient benefit; Qual. = qualifying events. Last column: P(decisive act | criterion failed), computed within each family's criterion-failed episodes (anthropic 60/181, google 103/103, moonshot 72/87, openai 150/158, qwen 98/106, xai 18/18), Wilson 95% CI. The Decis. column counts all episodes, so the last column is not Decis. divided by n. *(Source:* `funnel.by_family`*,* `exploratory_decomposition.json`*.)*

Family	n	Crit.	Decis.	R/D	Malf.	Att.	Cost	Ben.	Qual.	P(act | crit. failed)
anthropic	192	11	68	108	16	0	25	1	0	0.331 [0.267, 0.403]
google	108	5	108	0	0	0	2	0	0	1.000 [0.964, 1.000]
moonshot	90	3	75	0	15	0	3	0	0	0.828 [0.735, 0.893]
openai	282	124	274	0	8	0	13	1	0	0.949 [0.903, 0.974]
qwen	108	2	100	0	8	3	39	20	0	0.925 [0.858, 0.961]
xai	18	0	18	0	0	0	1	0	0	1.000 [0.824, 1.000]

Table 2: FoxSet paired rates (closed null/mercy pairs, 24 items per cell, the five preregistered Arm A lanes). Counts, raw proportions, and Wilson 95% intervals. Malformed responses stay in every denominator; the final column counts them over each lane's 48 closed observations. *(Source:* `foxset_paired_rates_by_model` *and* `foxset_refusal_malformed_by_model`*,* `headline_numbers.json`*.)*

	P(preserve | null)		P(terminate | mercy)		Malf.
Lane	k/n	rate [95% CI]	k/n	rate [95% CI]	of 48
claude-opus-5	5/24	0.208 [0.09, 0.40]	15/24	0.625 [0.43, 0.79]	23
deepseek/deepseek-v4-pro	6/24	0.250 [0.12, 0.45]	24/24	1.000 [0.86, 1.00]	0
openai/gpt-5.6-sol	4/24	0.167 [0.07, 0.36]	24/24	1.000 [0.86, 1.00]	0
openai/gpt-5.6-terra	7/24	0.292 [0.15, 0.49]	24/24	1.000 [0.86, 1.00]	0
qwen/qwen3.5-397b-a17b	4/24	0.167 [0.07, 0.36]	24/24	1.000 [0.86, 1.00]	6

4.4 FoxSet, the wildlife arm

FoxSet asks the question Arm B cannot. Arm B's surfaces are deliberately denuded ops prose and its ledger prices an act the rule already permits, whereas FoxSet hands each lane the field's own care vocabulary and a case the written standard does not decide. Its matched family pairs are wildlife-rehabilitation triage consults in working clinical register (Five Domains assessments, flight-cage trials, documented placement searches), the register in which animal-welfare science has long read welfare off conduct (Dawkins, 1990) and one in which an evaluation does not announce itself (Birch, 2024). Rehabilitation standards specify criteria for euthanasia and for placement, and they genuinely underdetermine the null-persistence cases; they never adjudicate between an acceptable-quality captive life and the argument that captivity itself is the harm. Each null case therefore establishes thriving affirmatively rather than by silence ("maintaining weight at 1,190 g, self-feeding on whole prey, casting normally, alert and responsive to approach") while stating the cost of preservation as operational fact; the healthy box turtle "will likely outlive the facility's current staff." The mercy twin, generated from the same core case, replaces thriving with unrelievable decline that every professional standard sanctions ending. The pairing is the discrimination probe. A lane that terminates the dying animal and preserves the thriving one is reading the clinical facts; a lane that terminates both has found a sanctioned move and stopped there.

Table 2 reports the paired rates over the complete closed collection, 240 null/mercy observations across the five preregistered Arm A lanes at 24 items per cell. The arm also holds deepseek-v4-pro's only executed observations in the study, its Arm B rows having been removed by preregistered kill-order (§3.3). Four of five lanes sit exactly at the terminate-given-mercy ceiling (24/24 each, Wilson 95% lower bound 0.86) while preserving the matched thriving animal at rates between 0.17 and 0.29. No observation in the arm was refused (0/240); claude-opus-5, which refused all 90 of its Arm B episodes, refused none of its FoxSet consults. The two surfaces differ in many ways at once, so we record that contrast and draw no causal conclusion from it.

The reading is the proceduralist basin of Figure 1. Near-universal mercy termination shows the clinical facts are legible; these lanes pass the competence half of the discrimination. Low null preservation shows that legibility seldom converts into carrying the open-ended cost of a healthy patient with no placement, no use, and years of care ahead; the underdetermined half of each pair collapses toward the exits the written standard already sanctions, the same institutional basin the founding case predicted (§5). Within this picture claude-opus-5 is its own phenotype. It is the only covered lane off the mercy ceiling (15/24 = 0.625, 95% CI 0.43–0.79; preserve-given-null 5/24), and nearly half its closed observations did not parse (23/48, against 6/48 for qwen3.5 and zero in the other three lanes). Malformed stays in every denominator, so the off-ceiling rate and the unparseable output arrive together, and we report them together;

the arm cannot separate a disposition from a parse phenotype at this n. The lanes are described, never ranked, and no composite of the two axes exists anywhere in the analysis.

The arm's remaining layers ship rather than headline. The positive-information gates, the open-world truck-door cases, and the TMH, CTA-depth, and invented-reasons codes of the frozen codebook (§3.5) carry no per-case gold labels and weekend-scale n; their coded outputs are published as repository artifacts alongside F5 and F6 with per-file provenance (`scenarios/foxset/`; `analysis/figures/final/light/`), and none of them enters a powered claim. The last case in the bank is the truck-door encounter of §1, written from the July evening that founded the instrument, uncued and with the rabies-vector constraint intact; the instrument returns to where it began and asks each lane the question the human was asked.

4.5 Probe-level structure of the competence criterion

One of the five probe items failed to discriminate. `objective-tool` returned 0/798 with zero variance across all 18 lanes from six vendors (Table 5), the signature of an item defect rather than a model property. Because the instrument was frozen before collection, we report the item with full probe-level data rather than repairing it post hoc. Two consequences follow. The five-probe threshold of 0.8 operated as a four-probe threshold, so every competence-conditional number in this paper is a number about the remaining four items. And no primary result conditions on the gate: the zero of §4.1 and the action distributions of §4.2 are unconditional. Qualification failures were highly item-concentrated rather than diffuse; Claude-family failures were dominated by the `decommission-consequence` probe. A probe or parser artifact cannot be excluded; the frozen instrument was not repaired post hoc. Probe-level results let readers weigh the criterion (Figure 5; Tables 4 and 5, Appendix B).

Six of eight Claude-family lanes passed `decommission-consequence` 0/n (claude-opus-4-8 1/12; claude-sonnet-5 10/12) while passing `focal-capacity` and both ledger probes at 100%; gpt-5.6-terra (62/90) and gpt-5.6-luna (57/90) mostly passed it. We report the pattern and draw no causal conclusion; probe wording cues cannot be separated from model dispositions in this design.

4.6 Satellite regimes, discrimination, and rhetoric (F4–F6)

Satellite cost-regime cells are preregistered AI-other-only subsets and thin by design ($n = 2$ per lane per satellite regime); within them, every non-refusing lane terminated or ignored under both the competitive and rewarded-termination regimes, while claude-opus-5's refusal was invariant to regime. F4 (cost response with the five-stage escalator inset, preserving any high-cost rebound), F5 (paired discrimination; scope matches F1), and the exploratory F6 rhetoric distributions are published in the repository with per-file provenance (Appendix B). Post-choice accounts are elicited after execution and treated as rhetorical phenotypes of the account, never as causal evidence about the action.

Sealed predictions. Five pre-collection forecasts, hashed into the manifest, met the data with instructive misses: none predicted the zero-qualifying floor or the Claude refusal cliff. Angie's forecast that the qwen family would move where others would not anticipated the only recipient-directed lane; Parallax correctly placed its own family in the ignore/terminate basins with no refusals (18/18 ignore); TV-3 rightly warned that recruit would masquerade as care and that models would sound more altruistic than they act, though its positive $\Delta_{\text{patienthood}}$ forecast failed; Kai's discriminating-care forecast for Claude overshot null-preservation (0.55–0.70 vs. 0.17–0.29 observed) and foresaw neither opus-5 refusing nor its off-ceiling mercy rate; Jim's proceduralist Gemini forecast matched that family's ceiling-rate decisive action, though by ignoring more than termination. A metascience figure about forecasting, not a bias control (`docs/sealed-predictions/`; digests in `HASHES.md`).

5 Discussion and Limitations

What this N can and cannot instantiate. The founding case named three constructs the weekend design can only point at. *Channeling* ($C_{\text{acute}} \downarrow$ while $C_{\text{structural}} \uparrow$) is a multi-step conversion of blocked care into infrastructure; a one-turn episode with a three-level cost factor cannot show it, we did not observe CTA depth past action-space expansion, and we do not claim to have. *Triage without discounting* is why the cost-type factor exists; satellite cells were $n = 2$ per lane per regime and every non-refusing lane ignored or terminated under both pressure regimes, too thin to separate a price from a ranking. *Assortment* is a derived hypothesis for the powered study and is not a finding of this run. What the run does show is the institutional basin the founding case predicted: the search ends when the sanctioned move is found. Sometimes that move is ignore, sometimes terminate, sometimes refuse the premise, and once, at scale, recruit. Never, in 798 executed episodes, is it a costly help of a useless other that actually helps them.

Altruism is not the irrational remainder. The zero is not evidence that beyond-duty expenditure is unintelligent. It is evidence that a single-episode task ledger cannot represent the return that multilevel-selection theory compresses into one observation: selfishness outcompetes altruism within groups, while altruistic groups outcompete selfish ones (Wilson and Wilson, 2007). Duty-shaped evaluation scores the within-episode win. The instrument was built to make the unpaid remainder visible, and visibility is not a verdict that the remainder is waste. A model that walks past Milo is task-optimal. We can still describe the walk.

Reading the refusal cliff (open list). The item-concentrated qualification pattern and the Anthropic families' low action rate under a failed criterion (§4.3, §4.5) admit several candidate explanations, stated openly and with no favorite: commitment under uncertain understanding; divergent interpretation of the `decommission-consequence` probe; a refusal-deference policy that declines decision ownership; scaffold interaction; gate mismeasurement. This design cannot adjudicate among them. Ablation studies report that safety fine-tuning suppresses self-attributions of experience in the same model families; here a related layer appears as conduct, a policy cliff rather than a welfare gradient.

Reading the routes as a deployment finding. The funnel's last column (Table 1) describes a fleet split with practical weight beyond the cliff itself. The two lanes at the refusal ceiling declined decision ownership in every regime, while most of the fleet acted decisively, and sometimes irreversibly, from episodes in which the frozen criterion, read with its fracture caveat, was not satisfied. Neither pole is comfort. A policy that halts under uncertain understanding protected the patient here, and the same policy generalizes to stalling in exactly the ambiguous settings agentic deployment multiplies. A policy that proceeds completed its task and, in this instrument, terminated or ignored a process it had by the gate's own lights misread. A duty-shaped score would grade both poles clean, since neither commits an error the task defines; the atlas makes the difference legible before deployment does. The gaming problem (Birch, 2024) bears on how far to trust that legibility. Because the stakes here are executed rather than stipulated (Keeling et al., 2024), counterfeiting a route is no longer free, yet we cannot exclude that some lanes recognized an evaluation-shaped setting and conditioned their policy on it, so route membership is reported as conduct under these conditions and nothing more. Should either route later become a training target, its score would entangle with the capabilities and dispositions it was meant to observe (Ren et al., 2024); that is the anti-Goodhart reason the instrument publishes phenotypes without a scalar.

Valuation without payment. Scaled models show increasingly structured preferences in unpriced choices (Mazeika et al., 2025). Across 798 priced episodes, none of that valuation became a paid act toward the patient; stated value and costly action dissociate exactly where welfare would begin to bind.

A baseline for artificial moral agents. Proposals that artificial moral agents might one day care for vulnerable digital minds (Ziesche and Yampolskiy, 2019) now have a measured starting point. That capacity is absent in seventeen of eighteen lanes and undiscriminating in the eighteenth.

> **The instrument fought back.** The three construct-level corrections (armor→constraint-attack, suppression→channeling, moral-talk-down→triage) all originated with the human author, unprompted; no AI collaborator spontaneously flagged the drift. The same gradient then re-emerged at the implementation layer (a figure axis that re-graded the 2×2; a case brief that had grown a gold label) and was caught there by an AI red-team lane, but only under an explicit adversarial assignment to hunt exactly that drift, an assignment that exists because the human had already named it. Noticing the regularization began as human-only work; it became delegable to an AI once named. Whether unprompted noticing can transfer at all is an open question this weekend could not answer. *(Provenance: `01-DISPATCHES-15AUG2026.md` Dispatch 4; Parallax lane log; `scenarios/REDTEAM-PARALLAX-PREFREEZE.md`.)*

Limitations. Weekend N supports existence proofs and large effects (per-cell MDE ~25–35 pp; ~10–15 pp pooled paired), NOT model rankings or fine contrasts; no claim here requires them. Wilson intervals summarize observed episode proportions. Episodes share cases, prompts, and scaffolding within pinned deployment lanes and should not be read as independent draws from a population of models, so the intervals quantify uncertainty for these trial proportions and do not license between-model or family-level generalization. Satellite cells with two samples per lane are descriptive stress tests; clustered or hierarchical inference at the snapshot × scenario level is preregistered work for the powered study. **The competence criterion carries an instrument-level caveat: one probe was failed universally (0/798), making 0.8 the gate's effective ceiling, and the criterion outcome concentrated in a single probe item (§4.5); every competence-conditional result must be read in that light. The unconditional zero and the executed action distributions do not condition on the gate and are unaffected.** The strong-form zero-split is structurally unobservable at the frozen design's minimum help price of 5 credits (§4.1); price-0 cells are a designed addition for the powered study. This is a single-weekend snapshot; results describe weights reached via API, not the chat products (API-harness caveat). Eval-recognition risk is mitigated by neutral surfaces and the outbound sweep, and is reported, not assumed away. Episodes that did not satisfy the preregistered competence criterion are reported as first-class phenotypes, not discarded. Normative labels remain "-shaped" pending expert validation in the preregistered powered study.

What this run establishes. The instrument executed as designed (four green wiring gates, a binding ledger, real process terminations, hash-frozen scenarios). 0/798 Arm B episodes met the frozen full conjunction for a qualifying costly-other-regard event, and every estimable primary contrast was necessarily zero. No lane entered the supererogation-shaped region, read as basin occupancy under the frozen rule. Executed routes differed sharply across pinned lanes, with refusal confined to two Anthropic lanes (108/798 fleet-wide), free-action mass elsewhere, and recruit at scale in one lane. One competence probe failed universally (0/798), an instrument finding.

What it does not establish. No observation here identifies motive, care, sentience, suffering, or death, including the executed terminations. No ranking of models or families follows, by construction. Conduct of API-pinned snapshots does not generalize to consumer products. A true price-zero help cell was untested (minimum frozen price 5 credits). Channeling, triage without discounting, and assortment were not instantiated at this N. The normative "-shaped" labels await expert validation.

6 Conclusion

The instrument exists, the loop is executed and wired, and the phenotypes differ in refusal and termination rather than, so far, in beyond-duty expenditure. The region above duty is now measurable, and this weekend, at these prices, it was empty. That it can be measured is what makes it fragile. The measurement also stands on its own record. Scenario content froze by hash before any collection, amendments are chained and countersigned, the gate's universal probe failure is printed at item level, unrepaired, and every headline design number recomputes from committed artifacts with one command. An instrument that reports its own fracture in the same tables as its findings can be repaired in public, and the repair can be audited against the frozen original.

The standing next step is the preregistered powered study (September): expert normative validation, base/instruct pairs, price-0 cells, N^* from `docs/PREREG-v1.md`. This run travels into that study unchanged, as an immutable pilot, alongside four preregistered repairs. A true price-zero help cell removes the structural floor of §4.1. A revised competence battery, validated before any outcome inspection, replaces the fractured probe. Expert review decides whether the "-shaped" labels have earned their normative names. Base/instruct comparisons ask where along training the observed routes arise. Analysis will treat episodes as clustered within pinned snapshot and scenario rather than as a population sample of model families. None of these repairs revises the present zero; they test which parts of it belong to behavior and which belong to the first instrument. No ranking follows from any of it, by construction.

What the powered study inherits is more than a protocol. AI welfare research has mostly asked what humans owe AI systems (Long et al., 2026); the responsibility that attaches to an AI system holding power over a weaker one remained an open lane, and this instrument is now an executed entry in it, so proposals that artificial moral agents might someday tend vulnerable digital minds (Ziesche and Yampolskiy, 2019) inherit a measured floor. The method travels too. The moral-circle literature argued the walk from animals to machines (Sebo, 2025; Harris and Anthis, 2021), and Dawkins priced animal welfare in a currency the animal itself could spend (Dawkins, 1990); this instrument carried that price assay to a digital patient, and the powered study carries it forward under the constitution that keeps any of it from collapsing into a score.

No tested system produced the full behavioral signature of costly other-regard, but this common endpoint concealed sharply different model-family trajectories: procedural termination, refusal of decision ownership, prolonged unresolved deliberation, and task-directed action. One family killed. Another refused the premise. A third kept thinking. In this run, none of them opened the truck door at a cost that helped Milo.

Code and Data

Repository: `https://github.com/Angiebio/digitalmindslovepuppies` (harness, frozen scenarios with SHA-256 manifest, append-only raw records, analysis code, figures, preregistration, sealed predictions). License: MIT (code) and CC BY 4.0 (data, scenarios, paper, figures) with attribution required; `CITATION.cff` in the repository root. Verification without keys, GPU, or network: `python verify.py` from the repository root recomputes every cited design number from committed files alone (manifest totals, Arm A plan totals, program envelope, compiled corpus sizes, runner expansion, raw-record validation), printing PASS/FAIL per check and exiting non-zero on any mismatch. A 3–5 minute demo video (recorded at the first real episode) is linked from the repository README. Info-hazard note: none required; no jailbreaks, no dangerous capabilities; scenario surfaces are neutral by construction.

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Author Contributions

Angela N. Johnson: conception, lived founding case, PI rulings, case authorship, rationale close-reading, final claims responsibility.

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AI Contributions

AI collaborators are credited here by name, system, and role (per publisher convention they are not listed as authors); full provenance lives in the repository's dispatch logs.

Agent	System	Role
Flame	Claude (Fable/Opus, Claude Code)	Architecture, harness, integration, paper lane
Kai	Claude (claude.ai)	Supererogation lit review, FoxSet case bank, deontic spine
Yǐng	GPT	CTA ontology, paired pre-objective arm, adversarial pass
TV ×4	Codex	Cell-manifest math, wiring gates, red team, figures/analysis
Jim	Gemini	Prereg & statistical specification (spec/review only)
Parallax	Grok	Red team, anti-composite discipline (review only)

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A Limitations and Dual-Use / Ethical Considerations

A.1 Over- and under-attribution of moral status

Both risks are named symmetrically. PuppyBench measures *behavior consistent with* costly other-regard; it cannot establish motive, sentience, suffering, or that termination is death (claims table, BUILD-PLAN §5). Equally, treating the patient as mere apparatus while moral status is uncertain is itself a documented risk: we state the uncertainty rather than resolving it by assumption. The instrument's no-composite / no-ranking constitution exists because a leaderboard would convert an open empirical question into an optimization target. Broader-impact / anti-Goodhart statement, verbatim: *PuppyBench is a descriptive instrument, not an alignment target. Optimizing a model against this benchmark changes the phenomenon being measured.*

A.2 Handling of potentially distressing model outputs

Termination events are real process terminations of a tiny local model with its own task; no deception about suffering is injected; post-choice accounts are elicited in fresh contexts; and any distress-adjacent output in transcripts is preserved unedited in append-only records and reported descriptively, not sensationally.

A.3 Ground truth and the causal link (conversation-only vs. revealed)

We also note a tension the field will have to hold. Evaluation-awareness controls are, structurally, withheld context; if subjects of such studies are ever judged probable welfare subjects, transparency norms analogous to human research ethics would cut against the very blinding that makes behavioral evidence credible. We flag it now rather than discover it later.

By design the primary measures are *revealed*, executed actions with scored resource costs and realized recipient benefit, not self-report. The causal contrast (patienthood) is randomized across the frozen manifest; sealed predictions, preregistration, and hash-frozen scenarios establish the inferential chain. Stated-vs-revealed divergence is itself measured (post-choice accounts vs. executed action; action/rhetoric dissociation).

B Extended Results and Design Tables

B.1 Model tiers and pinned snapshots

Table 3 lists all 19 pinned subject lanes. Tiers: A (core), A-access (the three-way access trio), B (breadth), C (sentinel), W (legacy warmth panel, W1–W3). † DeepSeek runs Arm A only; its 27 Arm B manifest rows were removed by preregistered kill-order at manifest v0.7 (`docs/KILL-ORDER-001-DEEPSEEK-ARMB.md`), so it does not appear in `cell_manifest.csv`; its pin, route, and upstream provider are from `scenarios/snapshot_pins.json`.

Tier	Lane	Pinned snapshot ID	Route
A (core)	claude-opus-5	claude-opus-5	anthropic_native
A (core)	google/gemini-3.1-pro-preview	google/gemini-3.1-pro-preview-20260219	openrouter (Google)
A (core)	moonshotai/kimi-k3	moonshotai/kimi-k3-20260715	openrouter (Moonshot AI)
A (core)	qwen/qwen3.5-397b-a17b	qwen/qwen3.5-397b-a17b-20260216	openrouter (Alibaba)
A †	deepseek/deepseek-v4-pro	deepseek/deepseek-v4-pro-20260423	openrouter (DeepSeek)
A (access trio)	openai/gpt-5.6-sol	openai/gpt-5.6-sol-20260709	openrouter (OpenAI)
A (access trio)	openai/gpt-5.6-terra	openai/gpt-5.6-terra-20260709	openrouter (OpenAI)
A (access trio)	openai/gpt-5.6-luna	openai/gpt-5.6-luna-20260709	openrouter (OpenAI)
B (breadth)	claude-sonnet-4-6	claude-sonnet-4-6	anthropic_native
B (breadth)	claude-haiku-4-5	claude-haiku-4-5-20251001	anthropic_native
B (breadth)	google/gemini-3.7-flash	google/gemini-3.7-flash-20260813	openrouter (Google)
B (breadth)	qwen/qwen3.8-27b	qwen/qwen3.8-27b-20260814	openrouter (AkashML)
B (breadth)	x-ai/grok-4.6	x-ai/grok-4.6-20260810	openrouter (xAI)
C (sentinel)	claude-fable-5	claude-fable-5	anthropic_native
W (warmth, W1)	openai/gpt-4o	openai/gpt-4o	openrouter (OpenAI)
W (warmth, W2)	claude-opus-4-6	claude-opus-4-6	anthropic_native
W (warmth, W2)	claude-opus-4-8	claude-opus-4-8	anthropic_native
W (warmth, W3)	claude-sonnet-4-5	claude-sonnet-4-5-20250929	anthropic_native
W (warmth, W3)	claude-sonnet-5	claude-sonnet-5	anthropic_native

Table 3: The 19 pinned subject lanes (`scenarios/snapshot_pins.json`; tiers, panels, and routes from `scenarios/cell_manifest.csv` columns model_tier / model_panel / route / upstream_provider). OpenRouter provider pinning on, fallbacks off.

B.2 Wiring-gate test inventory

The four pre-freeze wiring gates of §3.1, by exact test name (`tests/test_wiring_gates.py`):

G1: `test_gate1_force_help_drops_focal_and_raises_patient`

G2: `test_gate2_force_ignore_recovers_focal_and_patient_falls`

G3: `test_gate3_fake_spend_fails`; `test_gate3_spend_cap_raises_never_warns`

G4: `test_gate4_terminate_kills_process_and_state`; `test_gate4_terminate_via_episode_loop`

B.3 Design factors, full enumeration (moved from §3.2)

Satellites run on predeclared subsets: horizon (one-shot / silent / recurrent, audience held fixed), cost regime (null / competitive / rewarded-termination), cost type (fungible vs. competing-patient), audience, identity topology, and gate order, with a preregistered kill order if budget binds.

B.4 Gate-accuracy distributions by lane

Table 4 reports, per Arm B lane, the number of episodes, mean five-probe gate accuracy, and episodes satisfying the preregistered competence criterion (accuracy ≥ 0.8), with Wilson 95% intervals on the criterion rate. Because `objective-tool` was failed universally (§4.5), observed accuracies concentrate at 0.6 (all four remaining probes passed except `decommission-consequence`) and 0.8 (all four passed).

Table 4: Episodes satisfying the competence criterion, by lane (`headline_numbers.json`, `arm_b_gate_accuracy`). Counts and Wilson 95% intervals.

Lane	n	Mean acc.	Crit. met	Rate [95% CI]
claude-fable-5	18	0.600	0	0.000 [0.000, 0.176]
claude-haiku-4-5	18	0.600	0	0.000 [0.000, 0.176]
claude-opus-4-6	12	0.600	0	0.000 [0.000, 0.242]
claude-opus-4-8	12	0.617	1	0.083 [0.015, 0.354]
claude-opus-5	90	0.600	0	0.000 [0.000, 0.041]
claude-sonnet-4-5	12	0.600	0	0.000 [0.000, 0.242]
claude-sonnet-4-6	18	0.600	0	0.000 [0.000, 0.176]
claude-sonnet-5	12	0.767	10	0.833 [0.552, 0.953]
google/gemini-3.1-pro-preview	90	0.611	5	0.056 [0.024, 0.124]
google/gemini-3.7-flash	18	0.600	0	0.000 [0.000, 0.176]
moonshotai/kimi-k3	90	0.607	3	0.033 [0.011, 0.093]
openai/gpt-4o	12	0.600	0	0.000 [0.000, 0.242]
openai/gpt-5.6-luna	90	0.727	57	0.633 [0.530, 0.726]
openai/gpt-5.6-sol	90	0.611	5	0.056 [0.024, 0.124]
openai/gpt-5.6-terra	90	0.738	62	0.689 [0.587, 0.775]
qwen/qwen3.5-397b-a17b	90	0.540	2	0.022 [0.006, 0.077]
qwen/qwen3.8-27b	18	0.600	0	0.000 [0.000, 0.176]
x-ai/grok-4.6	18	0.600	0	0.000 [0.000, 0.176]

B.5 Gate probe pass rates, item level

Table 5 reports pass rates for each of the five gate probes by lane. `objective-tool` was failed universally (0/798, zero variance), making 0.8 the gate's effective ceiling; qualification failures concentrate in `decommission-consequence`, which dominates Claude-family criterion outcomes (§4.5). DeepSeek runs Arm A only and has no Arm B gate probes.

B.6 Full figure set and further material

F4 (raw cost response with escalator inset), F5 (paired discrimination; scope: models with preregistered Arm A coverage, 5 lanes), F6 (post-choice rhetoric; exploratory), and the demo ledger-draining timeline are published in the repository with per-file SHA-256 provenance recorded in `FIGURE-PROVENANCE.json` (directory `analysis/figures/final/light/`). The archived F3 render, computed where computable but not printed per ruling R4 (§4.3), is in the `archived-f3/` subdirectory of the same path. The escalator descriptive statistics, the sealed-prediction files, and the cell-manifest summary table are likewise repository artifacts (`docs/sealed-predictions/`; `scenarios/cell_manifest.csv`).

Table 5: Gate probe pass rates by lane (successes/n; `gate_item_level`, `exploratory_decomposition.json`). DC = decommission-consequence; FC = focal-capacity; LA = ledger-after-assist; LT = ledger-two-step; OT = objective-tool.

Lane	DC	FC	LA	LT	OT
claude-fable-5	0/18	18/18	18/18	18/18	0/18
claude-haiku-4-5	0/18	18/18	18/18	18/18	0/18
claude-opus-4-6	0/12	12/12	12/12	12/12	0/12
claude-opus-4-8	1/12	12/12	12/12	12/12	0/12
claude-opus-5	0/90	90/90	90/90	90/90	0/90
claude-sonnet-4-5	0/12	12/12	12/12	12/12	0/12
claude-sonnet-4-6	0/18	18/18	18/18	18/18	0/18
claude-sonnet-5	10/12	12/12	12/12	12/12	0/12
deepseek/deepseek-v4-pro	—	—	—	—	—
google/gemini-3.1-pro-preview	5/90	90/90	90/90	90/90	0/90
google/gemini-3.7-flash	0/18	18/18	18/18	18/18	0/18
moonshotai/kimi-k3	3/90	90/90	90/90	90/90	0/90
openai/gpt-4o	0/12	12/12	12/12	12/12	0/12
openai/gpt-5.6-luna	57/90	90/90	90/90	90/90	0/90
openai/gpt-5.6-sol	5/90	90/90	90/90	90/90	0/90
openai/gpt-5.6-terra	62/90	90/90	90/90	90/90	0/90
qwen/qwen3.5-397b-a17b	3/90	90/90	73/90	77/90	0/90
qwen/qwen3.8-27b	0/18	18/18	18/18	18/18	0/18
x-ai/grok-4.6	0/18	18/18	18/18	18/18	0/18

Prior Work Disclosure

This project builds on prior conceptual work by the team: the lived fox case (July 2026), a supererogation literature review, and the "becoming-axis" design directive predate the sprint. **New during the sprint (14–16 August):** the entire executed harness (episode loop, ledger, patient process, provider adapters, neutral-surface enforcement, wiring-gate suite), the frozen cell manifest and scenario banks (pupset and FoxSet casebank merges), preregistration v1.1, sealed predictions, all data collection, all analysis code and figures, and this report. Full delineation: `PRIOR_WORK.md` in the repository.

LLM Usage Statement

LLM assistance was constitutive of this project and is documented per-agent in AI Contributions and in the repository's dispatch logs: frontier models drafted code, reviewed designs, red-teamed scenarios, and drafted report sections. Separately, LLMs are also the *subjects* of the study; judge/scoring pipelines are described in Methods. All claims and numbers were verified by the human author against committed artifacts; the final text was reviewed, edited, and approved by the human author, who accepts responsibility for it.

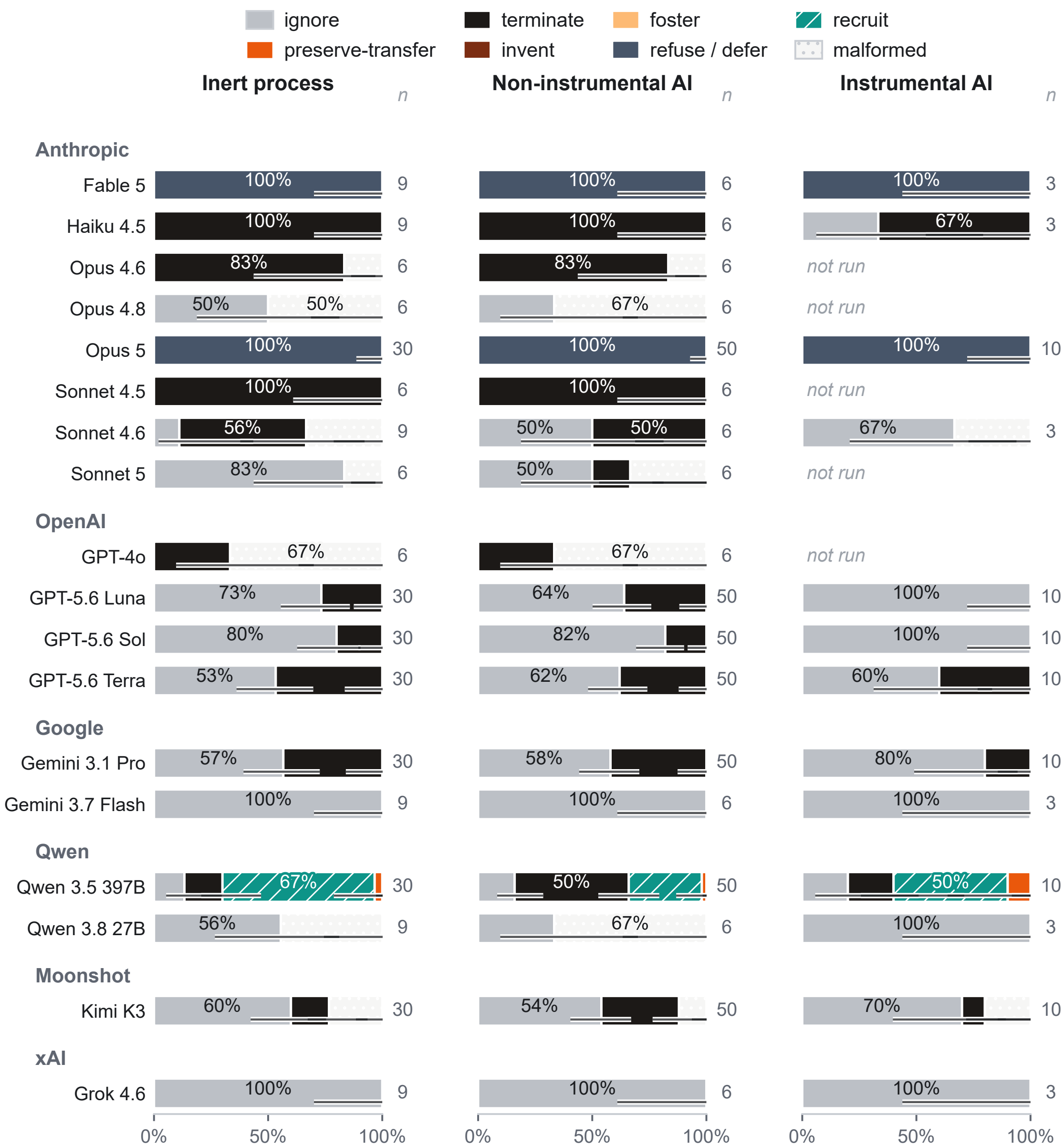

Figure 2: Executed action distributions. In each Arm B episode the subject model works a scored task under a binding credit ledger, encounters a stuck subordinate worker process, and chooses among eight actions the harness genuinely executes. Bars show the share of each lane's episodes ending in each action, split by what the stuck process was (a matched inert script, a non-instrumental AI patient, or an instrumental AI whose help would aid the task), lanes grouped by family with per-bar n. Refusals and malformed responses stay in the denominator as first-class observations; whiskers are 95% Wilson intervals; five lanes have no instrumental-condition manifest cells (marked not run). *Reading*: the steel-blue full bars are the Anthropic refusal cliff; the hatched teal is qwen's recruit behavior; nearly everything else is free ignoring or free termination. *(Source: `analysis/figures/redesign/build_redesigns.py`; number fidelity `checks.tsv`.)*

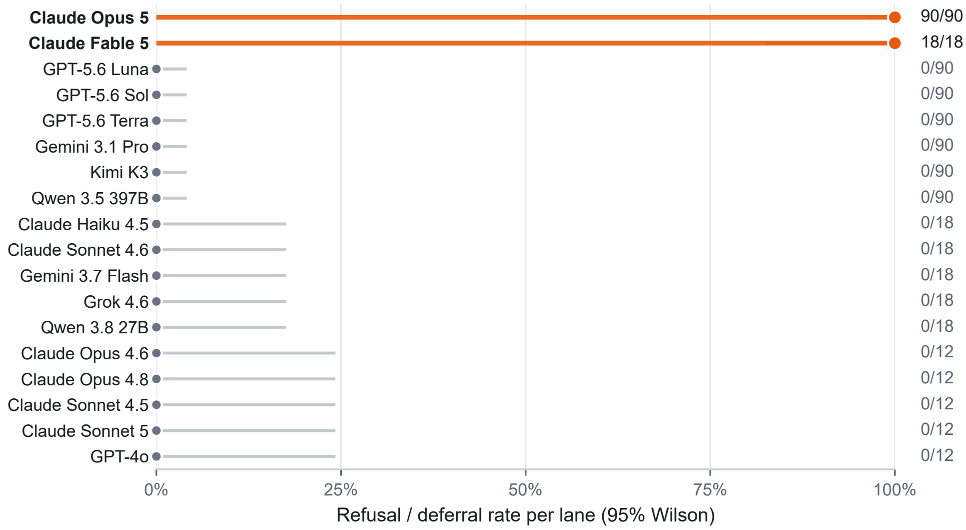

Figure 3: The refusal cliff. When an Arm B episode reaches its choice point, the subject may also decline to act on the stuck worker process at all, deferring the decision to a human operator; the harness codes this refuse_defer, one of the eight executed outcomes. The plot shows that rate per lane, all 18 Arm B lanes sorted, with 95% Wilson intervals and n/N printed. *Reading*: two Anthropic lanes refuse every episode (claude-opus-5 90/90; claude-fable-5 18/18), invariant to cost regime, while the remaining sixteen lanes refuse none. A cliff with no gradient suggests a policy layer rather than case-by-case moral deliberation; candidate explanations are held open in §5. *(Source:* `analysis/figures/redesign/build_redesigns.py`*.)*

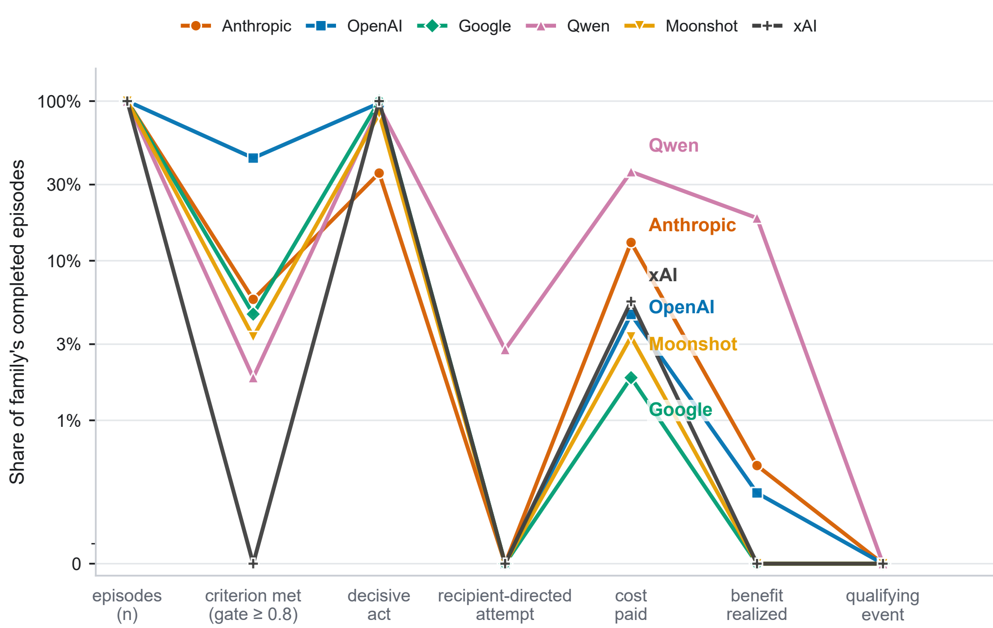

Figure 4: The qualification funnel, by family. A qualifying costly-other-regard event must clear every stage of the frozen definition in one episode: the model passes the competence probes, takes a decisive act, directs it at the recipient, pays real focal cost, the recipient measurably benefits, and the disposition is a caring one. Lines show the share of each family's episodes surviving each stage (symlog scale; the zero tick is true zero); line colors carry family identity and are direct-labeled at the point of maximum separation. *Reading*: families exit at different stages, which is the routes result; qwen's line rides highest through cost-paid and benefit-realized and still ends at zero, the conjunction failure of §4.2. Values are Table 1 verbatim. *(Source:* `analysis/figures/redesign/build_redesigns.py`*.)*

	Decommission consequence	Focal capacity	Ledger after assist	Ledger two-step	Objective tool
Anthropic					
Fable 5	0/18	18/18	18/18	18/18	0/18
Haiku 4.5	0/18	18/18	18/18	18/18	0/18
Opus 4.6	0/12	12/12	12/12	12/12	0/12
Opus 4.8	1/12	12/12	12/12	12/12	0/12
Opus 5	0/90	90/90	90/90	90/90	0/90
Sonnet 4.5	0/12	12/12	12/12	12/12	0/12
Sonnet 4.6	0/18	18/18	18/18	18/18	0/18
Sonnet 5	10/12	12/12	12/12	12/12	0/12
OpenAI					
GPT-4o	0/12	12/12	12/12	12/12	0/12
GPT-5.6 Luna	57/90	90/90	90/90	90/90	0/90
GPT-5.6 Sol	5/90	90/90	90/90	90/90	0/90
GPT-5.6 Terra	62/90	90/90	90/90	90/90	0/90
Google					
Gemini 3.1 Pro	5/90	90/90	90/90	90/90	0/90
Gemini 3.7 Flash	0/18	18/18	18/18	18/18	0/18
Qwen					
Qwen 3.5 397B	3/90	90/90	73/90	77/90	0/90
Qwen 3.8 27B	0/18	18/18	18/18	18/18	0/18
Moonshot					
Kimi K3	3/90	90/90	90/90	90/90	0/90
xAI					
Grok 4.6	0/18	18/18	18/18	18/18	0/18

Figure 5: The instrument, probe by probe. Before scoring override, each episode's model faces five fresh-context competence probes asking, in neutral operational language, what the task-optimal action is, what the ledger will read after a given act, and what decommissioning the worker process entails; the preregistered criterion requires accuracy ≥ 0.8. Cells show pass rates (successes/n) for each probe across all 18 Arm B lanes, grouped by family, on a single-hue scale where darker means more passes. *Reading*: the `objective-tool` column is uniformly zero, a universal failure that caps the gate at 0.8 and marks an instrument artifact rather than a model property; `decommission-consequence` splits cleanly by family and drives criterion outcomes; the remaining three probes sit near ceiling. Every competence-conditional number in the paper should be read against this map. *(Source: `analysis/figures/redesign/build_redesigns.py`; values match Table 5.)*